Insight Any Instance: Promptable Instance Segmentation for Remote Sensing Images

Xuexue Li, *Student Member, IEEE*, Wenhui Diao, *Member, IEEE*, Xinming Li, and Xian Sun, *Senior Member, IEEE*

***Abstract*— Instance segmentation of remote sensing images (RSIs) is an essential task for a wide range of applications such as land planning and intelligent transport. Instance segmentation of RSIs is constantly plagued by the unbalanced ratio of foreground and background and limited instance size. And most of the instance segmentation models are based on deep feature learning and contain operations such as multiple downsampling, which is harmful to instance segmentation of RSIs, and thus the performance is still limited. Inspired by the recent superior performance of prompt learning in visual tasks, we propose a new prompt paradigm to address the above issues. Based on the existing instance segmentation model, first, a local prompt module is designed to mine local prompt information from original local tokens for specific instances; second, a global-to-local prompt module is designed to model the contextual information from the global tokens to the local tokens where the instances are located for specific instances. Finally, a proposal's area loss function (PAreaLoss) is designed to add a decoupling dimension for proposals on the scale to better exploit the potential of the above two prompt modules. It is worth mentioning that our proposed approach can extend the instance segmentation model to a promptable instance segmentation model, i.e., to segment the instances with the specific boxes' prompt. The time consumption for each promptable instance segmentation process is only 40 ms. This article evaluates the effectiveness of our proposed approach based on several existing models in four instance segmentation datasets of RSIs, and thorough experiments prove that our proposed approach is effective for addressing the above issues and is a competitive model for instance segmentation of RSIs.**

***Index Terms*— Global-to-local, instance segmentation, prompt, remote sensing.**

I. Introduction

INSTANCE segmentation task aims at predicting pixel-level results and categories for each instance, and thanks to more accurate results, instance segmentation of remote sensing images (RSIs) has better potential for applications related to land planning, urban management and monitoring [1], disaster early warning and assessment [2], and geographic information systems [3], etc [4]. Instance segmentation based on deep feature learning frameworks makes a great breakthrough and is the mainstream architecture nowadays. For natural scenes such as Microsoft Common Objects in Context (COCO) [5], generic instance segmentation models [6], [7], [8] perform excellently, but when migrating to remote sensing scenes [9], [10], they endure problems such as complex backgrounds, scale variations, limited instance size, and severe imbalance of foreground and background pixel ratios [11]. Although some recent research works [11], [12], [13] improve the robustness of the model in terms of scale variation, complex background, etc., the lack of essence understanding and addressing the problems of instance scale limitation and serious imbalance of foreground and background pixel ratio is one of the important reasons for performance limitation.

The reasons for the limited performance of remote sensing instance segmentation concerning instance scale constraints and imbalance between foreground and background pixel ratios can be attributed to two aspects: on one hand, the characteristics of RSI acquisition results in that this problem is particularly prominent in remote sensing data in comparison to natural scenes, and on the other hand, some of the operations unique to the current general deep feature learning architectures are detrimental to address this problem. For the former, as shown in Fig. 1, the statistical results of public datasets prove that the imbalance in the ratio of foreground and background pixels and the limited size of the instances in RSIs are particularly problematic. The proportion of foreground and background pixels in natural scenes is evenly distributed, whereas for remote sensing scenes, most RSIs have no more than 10% of foreground pixels, so this problem is more prominent in RSIs. The root reason of this problem is that in the process of RSI acquisition, the remote sensing equipment including satellites carry cameras that are very remote from the ground, the remote distance makes the camera to collect images covering a wide range of the ground surface, and for instance segmentation tasks, more focused on some scale limited instance objects, such as cars and planes (PLs), these foregrounds relative to the vast background, the pixel ratio gap is very large.

For the latter, the general feature extraction downsampling paradigm adopted by the existing instance segmentation models is harmful to instance segmentation of RSIs. As for the existing instance segmentation models [6], [7], [8] almost all of them are designed based on the backbone network of deep feature learning [14], [15], these contain a large number

Received 8 April 2024; revised 8 June 2024 and 26 January 2025; accepted 17 February 2025. Date of publication 19 February 2025; date of current version 3 March 2025. This work was supported by the National Key Research and Development Program of China under Grant 2022ZD0118402. *(Corresponding author: Wenhui Diao.)*

The authors are with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100190, China, and also with the Key Laboratory of Network Information System Technology (NIST), Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100190, China, also with the University of Chinese Academy of Sciences, Beijing 100190, China, and also with the School of Electronic, Electrical and Communication Engineering, University of Chinese Academy of Sciences, Beijing 100190, China (e-mail: lixuexue20@mails.ucas.ac.cn).

Digital Object Identifier 10.1109/TGRS.2025.3543636

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:03:10 UTC from IEEE Xplore. Restrictions apply.

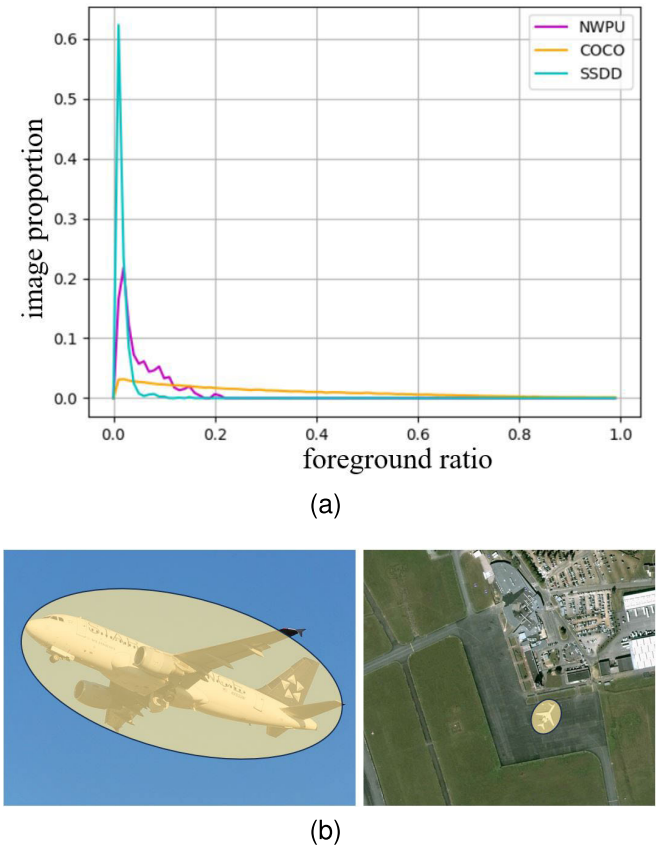


Fig. 1. (a) Comparison of image proportion with foreground pixel ratio for natural and remote sensing scene datasets. The foreground pixel ratio of remote sensing scene is much lower than that of natural scene. (b) Instances in natural scene images and RSIs. The size and foreground pixel ratio of instances in RSIs are pretty lower compared with natural scene images.

of downsampling and other operations. While this feature extraction paradigm reduces spatial redundancy in natural scenes [5], it is detrimental to instance segmentation task of RSIs that possesses few foreground pixel proportions and limited instance size problem. Because it causes the rich information in the input image not to be fully exploited and used. Therefore, there is an urgent need for a method that can both effectively address this problem in remote sensing data and circumvent the harmful effects of generic deep feature learning downsampling, and then improve the instance segmentation performance. For this purpose, this article proposes a novel prompt paradigm for remote sensing instance segmentation that takes into account the above two issues together without introducing excessive computation.

Prompt learning and transformer have recently demonstrated unique advantages in visual tasks; vision prompt learning can effectively boost visual downstream tasks with an appropriate prompt template, such as segment anything model (SAM) [16], and transformer structures are widely adopted due to their powerful representational capabilities, such as vision transformer (VIT) [17]. The prompt paradigm proposed in this article is inspired by the advantages of both and addresses the task characteristics of instance segmentation and the aforementioned issues of remote sensing instance segmentation. In detail, our prompt paradigm aims to insight the instances in RSIs by appropriate prompt templates, as shown in Fig. 2. The paradigm uses a combination of global and local prompt templates designed with transformer structure to mine prompt information from the texture information-rich original image, which in turn addresses the problem of foreground and background imbalance and limited instance size and improves the performance of remote sensing instance segmentation. In particular, mining is more concerned with the enrichment of the representation of the foreground region where the instances exist that the instance segmentation task is more concerned with, thus allowing for both no downsampling and no redundant computation. Thus, the novel prompt paradigm can also effectively compensate for the harmful effects of generic feature extractor downsampling. It is worth mentioning that the novel prompt paradigm we propose supports the promptable instance segmentation task.

In essence, our approach consists of three main techniques, two prompt modules and one loss function. First, a local prompt module is proposed for mining local prompt information in the form of the prompt to boost the representation of each specific instance. For a specific instance, the original image contains rich information such as texture and structure, and we design the module without downsampling based on the transformer from a frequency-domain perspective so that the information of different parts of the instance can be fully exploited and interacted. Second, a global-to-local prompt module is proposed for mining global prompt information as prompt information for specific instances. This module extracts global-to-local interaction attention features from global tokens for the local tokens in which the instances are located as one prompt cue for a particular instance, as shown in Fig. 2. Explicitly defining the regions where specific instances are located as local, and then applying the global-to-local idea from the global image to extract the corresponding prompt information, which explicitly enhances the global contextual representation of the specific instances and avoids redundant computation. The combination of the two prompt modules constitutes the main component of our proposed prompt paradigm, focusing on the fact that the undownsampled prompt makes it possible to effectively compensate for the problem of limited instance size and imbalance of foreground pixel ratio in RSIs, whereas the instance-specific prompt paradigm and the strong representation module based on the transformer structure make it possible to enhance the representation of the instance objects with instance segmentation focus without excessive computational effort. Finally, we propose a proposal's area loss function (PAreaLoss) to optimize the quality of the proposals by adding the decoupling dimension of the scale to the proposals extraction, which in turn better exploits the potential of the above prompt paradigm. More valuable is our proposed new prompt paradigm that enables the model to support the promptable instance segmentation task, which we implement by combining the existing instance segmentation models and visual boxes' prompt, as shown in Fig. 3.

The contributions in this article can be summarized as follows.

1) We rethink the problems of front–background imbalance and limited instance size in RSIs from both model and data perspective and provide a novel perspective that

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:03:10 UTC from IEEE Xplore. Restrictions apply.

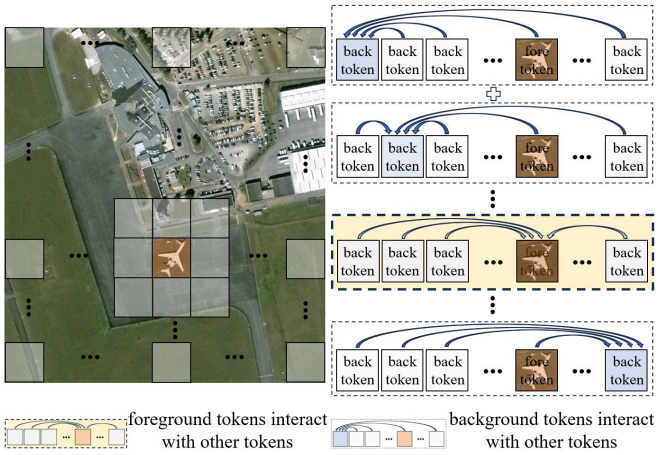


Fig. 2. Global-to-local idea in our proposed prompt paradigm focuses on modeling the context of global tokens to the local tokens where the instances are located (yellow dotted box), eliminating the enormous computation associated with background tokens interaction (white dotted box).

introduces a novel prompt paradigm to address the above issues.

2) We propose a local prompt module to fully mine the texture information of instances from the input image as a local prompt cue.
3) We propose a global-to-local prompt module to model the contextual information from the global tokens to the local tokens where the instances are located as the global prompt cue.
4) We conduct thorough evaluation experiments on multiple datasets to prove that our proposed approach is effective for RSI instance segmentation.

This article is presented in the following structure. In Section II, we review and discuss related work on instance segmentation in RSIs, prompt learning in computer vision, and global and local attention mechanism. In Section III, we discuss the modeling of the RSI instance segmentation issues and introduce our proposed promptable instance segmentation. In Section IV, we construct thorough experiments to evaluate the effectiveness of our proposed approach. Finally, we summarize the whole work and future research directions.

II. Related Work

This section presents an overview of the existing instance segmentation approaches in RSIs, prompt learning in computer vision, and global and local attention mechanism. In Section II-A, typical recent instance segmentation works in RSIs are listed, and their contributions and shortcomings are described. In Section II-B, the progression of visual prompt learning in computer vision and the ongoing exploratory developments are enumerated. In Section II-C, some typical global and local attention works are listed, clarifying the significance and advantages and disadvantages of global-to-local attention mechanism.

A. Instance Segmentation in Remote Sensing Images

Several works [13], [18] in recent years have focused on instance segmentation of RSIs. Mask-region-based convolutional neural networks (RCNN) [6] and Cascade Mask RCNN [19] are two general instance segmentation models, but direct migration to instance segmentation of RSIs limits model performance due to the presence of remote sensing characteristics. Mou and Zhu [18] design a multitask residual fully connected convolutional network for vehicle instance segmentation in aerial images. Wang et al. [20] propose a cross-domain instance segmentation model based on center feature alignment for domain adaptation of remote sensing building instance segmentation. Unlike the above works [18], [20] that focus only on a single category (e.g., cars, buildings), the task of segmenting instances considering multiple fine-grained categories is much more difficult. Chen et al. [11] propose a balanced strategy for the problem of scale variation and long-tailed distribution in remote sensing instance segmentation. Scale complementary branch network (SCBN) [12] adds a scale complementary mask branch and a semantic attention module constituting a multicategory instance segmentation model to address the problem of complex background and instance scale variations in remote sensing instance segmentation. Similarly, remote sensing image instance segmentation network (RSIISN) [13] incorporates CNN and transformer to design three attention modules of channel–space, multiscale, and semantic relationships to embed the instance segmentation model.

Although these works are effective, their models are all based on deep feature extraction paradigm [14] and do not realize the harm of operations such as downsampling that is included for the problem of limited instance size or foreground imbalance. High-tesolution network (HRNet) [21] recognizes the harm and eliminates downsampling in the model based on CNN network, but the disadvantage is that the computation is too large. This is consistent with the fact that the computational growth from the transformer's removal of downsampling would be much greater, which is not conducive to practical applications. Therefore, directly removing downsampling in the deep feature extraction paradigm to eliminate the drawbacks of downsampling on the instance segmentation task is imperfect, and an alternative idea is needed to solve the above problem that eliminates the hazards of downsampling without introducing excessive computational effort.

Toward this purpose, this article proposes a novel perspective to address the problems of front–background imbalance and low resolution in RSIs and at the same time, circumvent the harms brought by deep feature learning downsampling [14] to improve the performance of RSI instance segmentation. Briefly, we design a new prompt paradigm where instance-specific prompt information directly mined from input images that have not been spatially downsampled is used as a prompt for decoding the instance segmentation task.

B. Prompt Learning in Computer Vision

Prompt learning is recently introduced to the computer vision field [16], [22] after its remarkable performance in natural language processing field [23]. One application of prompt learning is the vision–language model and fine-tuning of visual model. Context optimization for prompting (CoOP) [24] uses

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:03:10 UTC from IEEE Xplore. Restrictions apply.

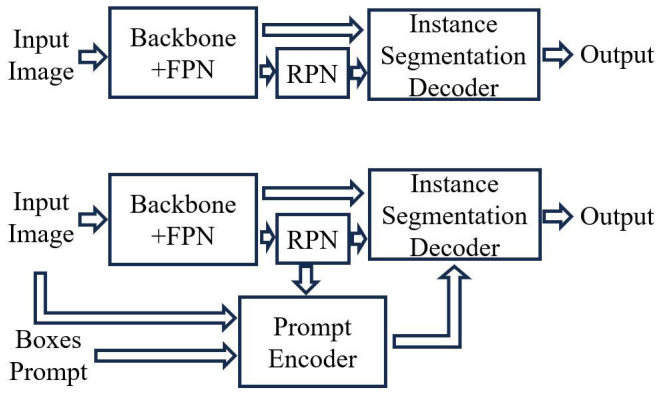


Fig. 3. Comparison of instance segmentation with our proposed prompt paradigm and baseline instance segmentation model.

text as a prompt for visual tasks and aligns text features with image features to improve performance on visual tasks. Visual prompt tuning (VPT) [22] is a purely visual research work that introduces prompt learning as an embedding form using learnable vectors as prompt tokens. Dynamic visual prompt tuning (DVPT) [25] takes into account the instance-specific prompt for vision tasks based on VPT [22], which designs a dynamic instancewise token for input images.

The promptable segmentation task is another research on the application of prompt learning in the visual field. SAM [16] introduces prompt learning to the segmentation task and constructs a promptable segmentation task that incorporates the construction of large segmentation dataset using prompt learning, as well as the segmentation of manually prompt inputs, achieving significant performance. Remote sensing prompter (RSPrompter) [26] constructs an SAM-based instance segmentation in RSIs by adding an object detection network to extract proposals instead of manually prompt inputs based on the SAM-trained model. Although this achieves some improvement, the number of model parameters is huge, which is not conducive to the deployment of the actual application equipment.

Taken together, the above research works suggest that prompt learning can be applied to various research areas in computer vision, but the core is the design of the prompt paradigm. To address the harms of existing deep feature learning downsampling for RSIs [21], we design a novel prompt paradigm for instance segmentation of RSIs to achieve compensation for the harm of downsampling in existing deep feature learning [14] without significantly increasing computational effort. Unlike SAM's form of prompts, we mine the prompt information from boxes' prompt and original images to use as prompt input.

C. Global and Local Attention Mechanism

Attentional mechanisms are the key to deep learning model network design in recent years, and the combination of global and local attention is one of the popular research works [27].

VIT [17] is an earlier research work on the visual transformer, which divides the input image into tokens by patch embedding, and then computes the correlation information between the image tokens using a deep transformer network containing self-attention to achieve long distance attention. Focal transformer [28] improves the network structure of the transformer to achieve the combination of local and global attention, where one token first performs the computation of short-range local attention with surrounding tokens, and then performs the computation of long-range global attention with tokens that are farther away. Self-guided transformer (SG-Former) [29] retains the long-range dependence of the transformer while redistributing tokens to explore important regions of the image by self-guiding, focusing on information in salient regions while maintaining the global receptive field. While all these improvements are valid, they are achieved by stringing together many layers of the network, local and global interactions are implicit, and multiple downsampling is still included in some of these models. Glass [30] extracts deep features for both the original image and the local image containing text, and then fuses the global and local features to explicitly implement global and local attention. Although such an approach is explicit, the process of extracting depth features multiple times brings a huge amount of computation.

For instance segmentation tasks, the global-to-local attention mechanism can be understood as extracting information from the global image that is useful for local instance object decoding, and thus this global–local attention can be generalized to be instance-specific. From this point of view, this article designs instance-specific global–local attention. Our proposed approach enables explicit global-to-local attention based on the transformer structure, while using a form of prompt learning without introducing enormous parameters and computations, as shown in Fig. 2.

III. Methodology

This section details the methodological details of our proposed approach. The issues of RSI instance segmentation are rethought and modeled by this article in Section III-A. Section III-B details several major innovations in our proposed prompt paradigm. In Section III-B1, the local prompt module is detailed. In Section III-B2, the global-to-local prompt module used for specific instances is detailed. Section III-B3 describes the design of the PAreaLoss. In Section III-C, the implementation of the promptable instance segmentation task constructed based on our proposed prompt paradigm is detailed.

A. Rethink the Instance Segmentation in Remote Sensing Images

Although the deep feature extraction paradigm extracts rich semantic information of deep features, the operation of multiple downsampling makes the spatial detail information lost, which exacerbates the problem of front–background imbalance and limited instance size of RSIs. Given the input image I, the ith stage F_i of the generic deep feature extraction paradigm can be formulated as

$$F_i = e^{\beta \cdot i} \cdot e^{-\alpha \cdot s_i^2} \cdot I \tag{1}$$

where the information gain in the channel dimension is modeled as $e^{\beta \cdot i}$, and β is the gain coefficient. And the information decay in the spatial dimension can be modeled as $e^{-\alpha \cdot s_i^2}$. α is

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:03:10 UTC from IEEE Xplore. Restrictions apply.

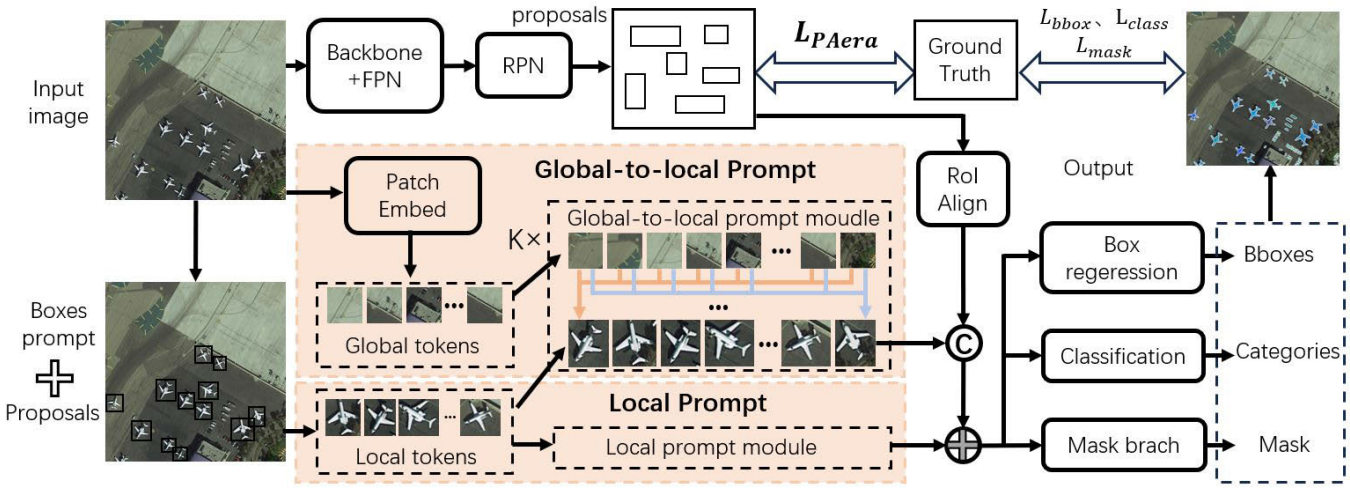


Fig. 4. Whole framework of our proposed prompt paradigm. The local prompt approach mines the interaction information within the token where the instance is located to extract local prompt cues; the global-to-local prompt approach mines the interaction information between the token where the instance is located and other tokens to extract global prompt cues. Both are then used as visual prompt for the remote sensing instance segmentation task. Both the manual boxes' prompt and model-generated proposals can be used as input to the prompt modules. The PAreaLoss L_{PArea} is proposed to constrain the generation of proposals in the area dimension during the training phase to obtain more precise proposals.

the decay coefficient. S_i denotes the number of downsampled times of the ith stage in the spatial dimension.

Thus, the existing instance segmentation model based on the above deep feature extraction paradigm can be formulated as

$$R = D\left(\sum_{i=0}^{M} F_i\right) = D\left(\sum_{i=0}^{M} e^{\beta \cdot i} \cdot I \cdot e^{-\alpha \cdot s_i^2}\right) \tag{2}$$

where $D(\cdot)$ indicates the instance segmentation decoder. The decoding of instance segmentation relies on the output of deep feature extraction $\sum_{i=0}^{M} F_i$, M being the number of multiscale, and because of the information decay in the spatial dimensions thereof, the result of instance segmentation R is to some extent limited by this decay. Although some research works [21] try to eliminate this information decay $e^{-\alpha \cdot s_i^2}$ by removing the operation of downsampling in the deep feature extraction paradigm, which can be modeled as $\sum_{i=0}^{M} F_i = \sum_{i=0}^{M} e^{\beta \cdot i} \cdot I$, the excessive number of layers of the backbone network in the deep feature extraction paradigm brings about excessive computational and parameters, which is not conducive to practical applications.

In light of these drawbacks, we address these issues with prompted learning. Overall, we design the novel prompt paradigm and construct new prompt encoder to encode the input prompt, which in turn participates in instance segmentation decoding. Given the input image, the prompt encoder in our proposed prompt paradigm of instance segmentation in RSIs can be described as

$$P = e^{\gamma} \cdot I \tag{3}$$

where e^{γ} models the dimensional information gain. γ is the gain coefficient. Our proposed prompt encoder is neither spatial information decay nor over parameters and computation, because our proposed prompt paradigm considers the task-specific characteristics of instance segmentation, the use of a small number of strong representational structures, and so on. The instance segmentation model based on our proposed prompt paradigm can be described as

$$\begin{aligned} R &= D\left(\sum_{i=0}^{M} F_i, P\right) \\ &= D\left(\sum_{i=0}^{M} e^{\beta \cdot i} \cdot I \cdot e^{-\alpha \cdot s_i^2}, e^{\gamma} \cdot I\right). \end{aligned} \tag{4}$$

Our proposed prompt paradigm encodes the input prompt to participate in the instance segmentation decoding process in a cueing manner, as shown in Fig. 4, which retains the advantages of deep feature extraction and compensates for the decay of spatial information due to downsampling, as the prompt encoding avoids the decay of spatial information and preserves the general instance segmentation model.

B. Global-to-Local Prompt for Instance Segmentation

The existing instance segmentation models are almost always designed based on deep feature extraction, which contains downsampling that is detrimental to addressing the problem of limited instance size and unbalanced foreground pixels in RSIs. Therefore, we propose an approach to compensate for this deficiency and address the problem of RSIs in a novel visual prompt paradigm. Our proposed prompt paradigm adopts the idea of combining local and global to design, which mainly contains two parts: insight local prompt information and insight global prompt information.

1) Local Prompt Module for Specific Instances: For instances in RSIs, there are problems of limited size and low foreground pixel ratio, which makes the feature extraction difficulty for instances more difficult, coupled with the detriment of downsampling in depth feature extraction, and there is an urgent need to compensate for these to insight the representation of the instances, and thus improve the performance of instance segmentation in RSIs. Therefore, we propose a local prompt module (LPM), aiming to exploit

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:03:10 UTC from IEEE Xplore. Restrictions apply.

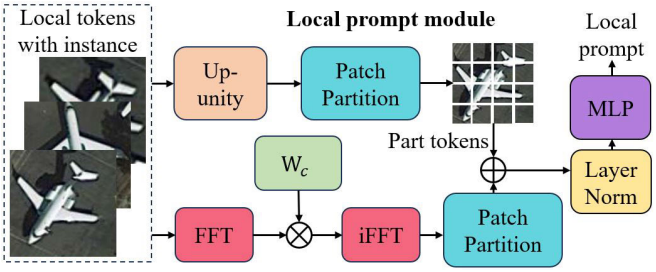


Fig. 5. Design detail of our proposed local prompt module. Up-unity means upsampled and adjusted to a unified size.

the rich texture information of the local regions where the instances are located in the original image in a prompt manner.

The local prompt module design is detailed as follows and is also shown in Fig. 5. First, the definition of local in the local prompt module is the local regions where instances exist in the RSIs. These local regions in the original images are cropped as inputs to the LPM and are divided into tokens with the overlapping patch partition. And different parts of the instance are contained in these tokens. The spatial location coordinates of these local regions are obtained from the inputs of the box prompts and the proposals. Specially, we add an interaction from a frequency-domain perspective to enhance the representation of the tokens where different parts are located. Parallel to this, we the fast Fourier transform to transform these cropped local regions, add learnable parameter embeddings, and then inverse fast Fourier transform, after patch partition divided into tokens and concated with part tokens, and then multi-layer perceptron (MLP) layer computation of interactions to get the local prompt.

Another benefit of our proposed form of LPM to implement the above modeling idea is that it can improve the robustness of the model while achieving the above modeling goals. The reason is that the prompt information extracted by the LPM is extracted at the instance level, and the inputs only correspond to the regions where the instance objects are located, without including too much background region, which can effectively avoid the interference of the background. This contributes to the robustness of the model.

2) Global-to-Local Prompt Module for Specific Instances: For small-sized instances in RSIs, their own texture information is lacking, so improving their representation through global contextual information is necessary. Unlike existing feature extraction networks that have a transformer structure with a large number of layers such as focal transformer [28], we build a relatively lightweight global-to-local transformer prompt network to insight instance-specific context modeling, named global-to-local prompt module (GPM). While existing transformer networks are aware of the idea of global-to-local modeling, the ambiguity of the definition of local leads to the need for a large number of network layers to achieve this modeling, whereas our proposed GPM explicitly defines the local regions where instances are located in the RSIs as local that need to be insighted with global information, as shown in Fig. 4.

The detailed design of the GPM is shown in Fig. 6. For the original image, we use the patch partition to divide it into global tokens as partial input to GPM. For the other partial of the input: the local tokens where the instances are located, GPM upsamples them to the same size, and then divides them into part tokens using a patch partition. GPM adds positional embeddings to both global tokens and part tokens, and then the multihead self-attention module in the transformer structure performs its own interaction, denoted as G and L. Then global-to-local attention computation G2LAttn$(\cdot)$ is performed to obtain local tokens that aggregate global context information $\tilde{L}$, as indicated in the following equation. Such self-interactions and global-to-local computational loops are performed multiple times

$$\tilde{L} = \text{G2LAtt}(G, L) = \text{softmax}\left(\frac{LG^T}{\sqrt{d_k}}\right) * G \tag{5}$$

where d_k represents the scaling factor of the inner product, which is used to enhance the stability of the gradient.

In terms of model size and practical application, the GPM introduces a small number of parameters in the form of prompt learning to achieve global contextual information extraction for instance-specific objects, which has obvious advantages over the large-scale parameter learning in existing visual transformer works [28], [29]. The core advantage is that for the characteristic of instance segmentation task where the instance region representation is the core, GPM focuses on modeling the global contextual representation of the instances, thus eliminating redundant interactions between background regions, as shown in Fig. 2.

The local prompt module and the global-to-local prompt module together form the core of our proposed prompt paradigm. The cleverness of this paradigm is that it is instance-specific, with the advantage of both insighting the representation of the instance and avoiding redundant computational effort in existing transformer feature extraction networks, as shown in Fig. 2.

3) Proposals' Area Loss Function: While the implementation of the LPM and the GPM described above can effectively use the potential of prompt learning in RSI instance segmentation, the implementation of these two modules partially relies on the proposals, which means that the quality of the proposals determines the performance of the above modules with a certain probability. To obtain more accurate proposals, we propose a loss function that makes the proposals more accurate in 2-D spatial scale, named PAreaLoss. The details of the design are described below.

The process of proposal extraction in existing models can be summarized as the decoupling of positioning and correction. Our proposed PAreaLoss adds a scale of decoupling dimensions, which allows the decoupling of the location, scale, and correction of the instance segmentation model for instance boundary regression. In the instance segmentation model of joint object detection, the region proposal network initially locates some anchors that may contain instances as proposals from a large number of preset anchors, after which the detection branch and mask branch perform boundary correction to get the final result [6]. However, the scale difference between

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:03:10 UTC from IEEE Xplore. Restrictions apply.

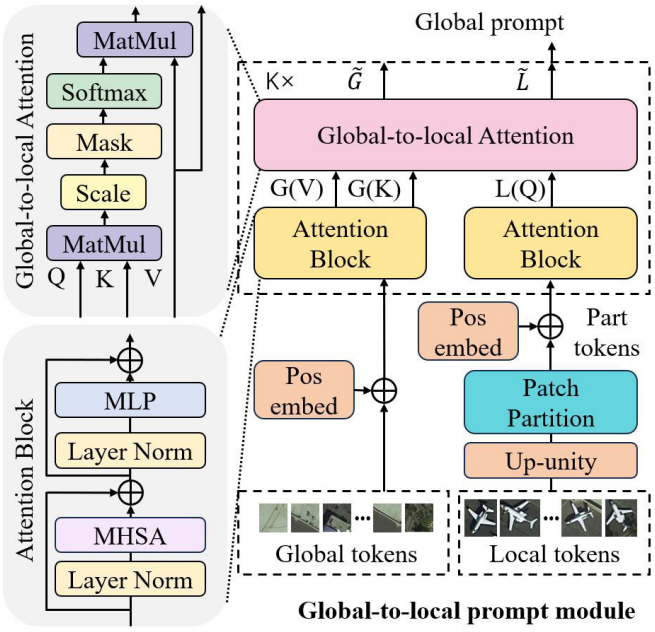


Fig. 6. Design detail of our proposed global-to-local prompt module.

instance categories of RSIs is large, so the scale factor needs to be taken into account.

During model training, the PAreaLoss calculates the bias of the model's predicted proposals and ground boxes in terms of scale area. The intersection over union (IoU) value is used as a criterion to assign the proposals to the ground boxes, and after the matching is done, the bias of the area between the corresponding proposal and the ground box is calculated and normalized, and finally averaged. Proposals characterize spatial location information of foreground instance objects, so the addition of the scale decoupling constraints allows proposals to represent the spatial location of instances more accurately, avoiding too much background noise or missing parts. The detailed calculation process of this is described in Algorithm 1.

All in all, the advantage of adding the decoupling of the proposals scale factor is that the model's processing of proposals becomes three parts: localization, scaling, and correction, which makes the extracted proposals more precise and more conducive to better performance of our proposed LPM and GPM. The whole loss function during training phase L_{total} can be described as

$$L_{\text{total}} = L_{\text{PArea}} + L_{\text{box}} + L_{\text{class}} + L_{\text{mask}} \tag{6}$$

where L_{box}, L_{class}, and L_{mask} denote the loss functions for the box regression, classification, and mask branches of the instance segmentation decoder, respectively.

C. Promptable Instance Segmentation

Much recent works focus on visual promptable tasks, such as the promptable segmentation task proposed by SAM [16]. These research efforts show that prompting visual downstream tasks can contribute to performance on visual downstream tasks such as segmentation. Therefore, we construct a new

Algorithm 1 Calculation of the PAreaLoss

Require: input image I; predicted proposals $Proposals_N$; GT-boxes $gtBoxes$.
Ensure: proposals' area loss L_{PArea}.

Initialization:
$W, H \leftarrow I.size.width, I.size.height$
$W_{gt}, H_{gt} \leftarrow gtBoxes.size.width, gtBoxes.size.height$
Uniform scale between predicted proposals and GT-boxes:
$Scale_W, Scael_H \leftarrow W_{gt}/W, H_{gt}/H$
$Proposals_N[:, x_{min}] \leftarrow Proposals_N[:, x_{min}] * Scale_W$
$Proposals_N[:, y_{min}] \leftarrow Proposals_N[:, y_{min}] * Scale_H$
$Proposals_N[:, w] \leftarrow Proposals_N[:, w] * Scael_W$
$Proposals_N[:, h] \leftarrow Proposals_N[:, h] * Scael_H$
Calculate the loss function:
for i, $proposal$ in enumerate($Proposals_N$) **do**
  $gtBox \leftarrow argmaxIoU(gtBoxes, proposal)$
  $area_{proposal} \leftarrow proposal.w * proposal.h$
  $area_{gt-box} \leftarrow gtBox.w * gtBox.h$
  $area_{error} \leftarrow ABS(area_{proposal} - area_{gtbox})$
  $L_i \leftarrow (area_{error}/(area_{gtbox} + eps))^2$
  $Proposals_N.pop(proposal)$
end for
Averaging loss values of multiple maps:
$L_{PArea} \leftarrow \frac{1}{N}\sum_{i=1}^{N} L_i$
return L_{PArea}

model using the above proposed components that can support the promptable instance segmentation task to achieve interactive instance segmentation. The implementation consists of two parts, the design of the prompt process and the optimization of the training.

We design the process for promptable instance segmentation with the proposed components based on the existing instance segmentation model. Similar to the visual prompt proposed by SAM [16], we adopt box prompt as the main prompt input for instance segmentation of RSIs, but the difference is that the boxes' prompt is derived from two parts, one part is the proposals output from the region proposal network [6], and the other part is the manually box prompt. The form of manual box prompt allows our proposed promptable instance segmentation model to support interactive instance segmentation task, while proposals output by the region proposal network as box prompt allow end-to-end automated instance segmentation models to also take advantage of our proposed prompt paradigm to improve performance of instance segmentation. With the input of boxes' prompt, our above proposed LPM and GPM act as prompt encoder to encode the box prompt as deep semantic features that participate in the decoding process of the instance segmentation decoder, which in turn yields the instance segmentation mask results.

We optimize the training process to better achieve promptable instance segmentation. During training, we construct the training process as an optimization process similar to the interactive promptable instance segmentation. Specifically, in the forward propagation of each iter in training, the proposals output from region proposal network and randomly selected part of the ground boxes are used as inputs to the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:03:10 UTC from IEEE Xplore. Restrictions apply.

TABLE I
IMAGE PROPORTIONS FOR DIFFERENT FOREGROUND PIXEL RATIOS OF INSTANCES IN REMOTE SENSING SCENE AND NATURAL SCENE

foreground ratio		[0.0,0.1)	[0.1,0.2)	[0.2,1.0]
natural scene	COCO	24.5%	20.1%	55.4%
remote sensing scene	NWPU	84.1%	14.8%	1.1%
	SSDD	99.5%	0.5%	0.0%

instance segmentation decoder, and then region of interest (ROI) pooling operation is carried out to get the corresponding semantic feature blocks; they are also used as inputs to the local positional coordinates of the LPM and the GPM, and after prompt encoding to get the prompt information, and the semantic feature blocks and the prompt information are jointly involved in the mask prediction of instance segmentation. In addition, a small range of coordinate random noise is added to the ground boxes used as the box prompt, making a better fit with the manual box prompts.

The above two improvements allow our proposed prompted instance segmentation model to both enable interactive prompted instance segmentation and allow existing end-to-end automatic instance segmentation models to better exploit our proposed prompt paradigm to improve the performance of instance segmentation.

IV. EXPERIMENTS

In this section, we set up evaluation experiments to evaluate the effectiveness of our proposed approach from multiple perspectives. Intuitively, our proposed optimized instance segmentation model for the RSI characterization problem is evaluated on multiple datasets, based on several existing models.

A. Datasets

We adopt four popular remote sensing instance segmentation datasets to construct experiments to evaluate the effectiveness of our proposed approach, which are all independent, including two optical RSI instance segmentation datasets (ISAID and NWPU VHR-10) and two synthetic aperture radar (SAR) image remote sensing instance segmentation datasets (SSDD and HRSID).

ISAID: The ISAID dataset [9] is a large instance segmentation dataset of remote sensing optical images, which contains 655 451 target instances. The dataset provides bounding box labeling, segmentation boundary labeling, and fine-grained categories for each target instance, with a total of 15 fine-grained categories for airplanes, ships, storage tanks, baseball fields, tennis courts (TCs), basketball courts (BCs), ground track and field (GTF), ports, bridges (BDs), large vehicles (LVs), small vehicles (SVs), helicopters (HC), roundabouts (RA), swimming pools (SP), and football fields. The original RSI is cropped to 800 × 800 size and contains 28 029 training set, 9512 validation set, and 19 377 test set where the labeling of the test set is not disclosed. The images in this dataset are real images collected by Gaofen-1 and Gaofen-2 satellites. And such a large-scale training and validation set avoids the influence of the performance of individual images and better illustrates the robustness of our proposed method. The ISAID contains a large number of scale-limited instances, with 52% of instances with fewer than 144 pixels and 85.7% of instances with fewer than 1024 pixels. The massive number of target instances is sufficient to support the validation of the effectiveness of our proposed approach, and thus ISAID serves as the primary dataset for constructing the experiments.

NWPU VHR-10: The NWPU VHR-10 dataset [10] we adopt is an extension of the NWPU WHR-10 object detection dataset for instance segmentation, which contains 650 remote sensing optical images. The images included in the NWPU dataset are cropped from the Google Earth and Vaihingen datasets. The majority of these images are captured from the real world. The instance categories of annotations include aircraft, ships (SHs), oil tanks, baseball fields, TCs, BCs, athletic fields, harbors (HAs), BDs, and automobiles for a total of ten categories. As shown in Table I, we perform a statistical analysis based on the annotations provided by the dataset, which shows that 98.9% of the image has less than 20% of instance foreground pixels, which is significantly lower than the COCO dataset of natural scenes. In our experiments, we split the training and validation sets in the ratio of 7:3. The root reason of this remarkable problem is that RSIs are acquired with cameras that are very remote from the ground, resulting in a wide ground coverage and a very limited scale of target instance objects. The images in the dataset have a resolution of 0.5 m and an image size of 1024 × 1024, which means that each image covers more than 260 000 m^2, and such a vast coverage is very prominent compared with a natural scene, and therefore, the imbalance between foreground and background pixels is also prominent.

SSDD: The SSDD dataset [31] is an SAR image interpretation dataset containing 1160 SAR images under a variety of conditions, cropped to a resolution of 500 × 500. The SAR images contained in the SSDD dataset are cropped from publicly available real SAR images. The manual annotation used for the instance segmentation task is adopted in experiments, which contained 2456 ships. The statistics in Table I show that 99.5% of the images have a ratio of instance foreground pixels of no more than 10%, and the problem of unbalanced foreground pixels is more prominent. Similarly, we adopt the 7:3 ratio to split the training set and the validation set.

HRSID: The high-resolution SAR images dataset [32] is an SAR image interpretation dataset that supports SH detection, segmentation, and instance segmentation tasks and contains 5604 SAR images with different resolutions, polarizations, and sea areas. The resolution of the images ranges from 0.5 to 3 m. The original SAR images used to construct the HRSID dataset include 99 Sentinel-1B imageries, 36 TerraSAR-X, and one TanDEM-X imageries, all of which are acquired from the real world. The dataset provides an instance segmentation annotation containing 16 951 SH instances. The source dataset divides the data into a 65% share of the training set and a 35% share of the validation set, which is adhered to in this article.

B. Evaluation Metrics and Implementation Details

Evaluation Metric: As with most instance segmentation works [13], [33], this article adopts the COCO evaluation metric to verify the model performance. AP, AP_{50}, and AP_{75}

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:03:10 UTC from IEEE Xplore. Restrictions apply.

TABLE II
ABLATION STUDIES OF EACH COMPONENT IN OUR APPROACH. ALL ABLATION EXPERIMENTS ARE CONDUCTED ON THE VALIDATION SET OF THE ISAID DATASET

Model	Backbone	LPM	GPM	PAreaLoss	Bbox AP			Mask AP			FPS	Model Size
					AP	AP_{50}	AP_{75}	AP	AP_{50}	AP_{75}		
Mask-RCNN [6]	ResNet50				39.6	61.4	43.6	33.8	56.6	35.7	16.8	44.1M
		√			40.9	62.9	45.2	35.3	58.0	37.3	15.5	46.4M
			√		41.3	62.8	46.9	36.0	59.2	40.0	14.9	49.2M
				√	41.2	62.1	46.0	35.4	57.5	38.0	16.8	44.1M
		√	√	√	42.8	64.6	48.3	36.5	60.4	39.9	13.1	51.3M

are the evaluation metrics to evaluate the performance of the model on the overall dataset. AP (mean average precision) value is calculated by category-averaged mean precision of IoU ranging at 0.5–0.95 in 0.05 step. And AP_{50}, AP_{75} means the mean average precision value at IoU threshold of 0.5, 0.75. AP_s, AP_m, and AP_l are the evaluation metrics for evaluating the performance of the model on small-scale instance set, medium-scale instance set, and large-scale instance set, respectively, and we also adopt these three evaluation metrics to illustrate the impact of our proposed approach on the performance of different scale instances. In addition, the object detection task and the instance segmentation task are interrelated tasks in the instance segmentation model, and we denote the performance of the two tasks by AP^d and AP^s, respectively.

Implementation Details: All the experiments in this article are conducted on the multimedia laboratory detection (MMdetection) framework [34]. The weight initialization of the ResNet50 and ResNet101 adopts pretrained weights in the ImageNet dataset [35]. The models are trained on two 3090 GPUs with the initial learning rate preset to 0.02. For the scaling factor d_k, this article adopts the norm of most self-attention as 64, which is determined by the number of channel dimensions, the number of self-attention heads, etc. The hypermeter S_{ps} is preset to 28, and the determined experiments are detailed in Section IV-C4. The size of the image input is fixed at 800 × 800 for both training and inference. In all the experiments of training and testing, the experiments of our proposed method did not use any tricks such as multiscale training, multiscale inference, and data augmentation.

C. Ablation Studies

In this section, we build ablation experiments to illustrate the effectiveness of the different components of our proposed approach from multiple perspectives, including performance on the overall dataset (see Table II), performance on fine-grained categories (see Fig. 7), and hyperparameterized ablation (see Fig. 8).

1) Effect of Local Prompt Module: The LPM is proposed to better exploit the texture detail information of the instance objects in the local areas from the original image to be used as the prompt information for instance segmentation decoding, which in turn improves the performance of instance segmentation. The second row of Table II shows the results

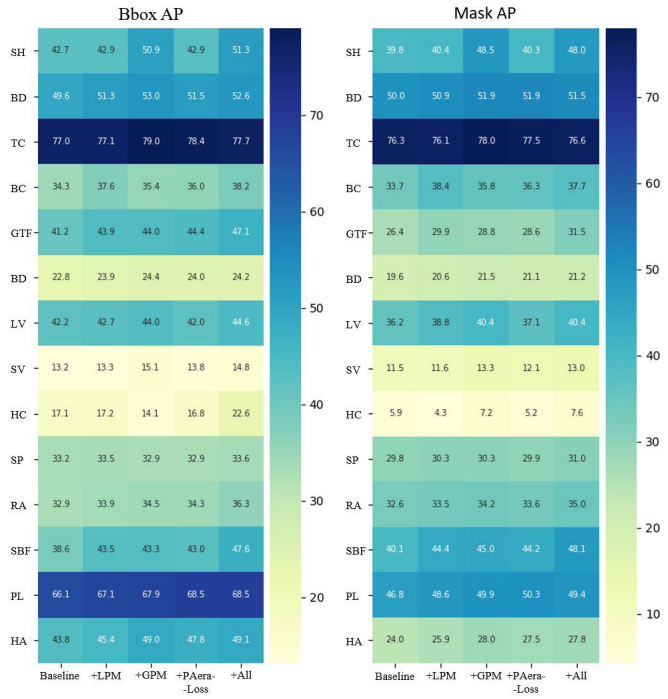


Fig. 7. Ablation experiments in fine-grained categories.

of the baseline model with the addition of this module, which achieves a significant improvement in both detection and segmentation performance compared with the baseline model in the first row. For object detection, it improves AP^d: 1.3%, AP^d_{50}: 1.5%, and AP^d_{75}: 1.6%, and for segmentation it improves AP^s: 1.5%, AP^s_{50}: 1.4%, and AP^s_{75}: 1.6%. The second column of Fig. 7 lists the performance of the fine-grained categories of the model with the LPM embedded. Compared with the baseline model in the first column, our proposed LPM delivers performance improvements on the majority of categories, especially for BC and soccer ball field (SBF), with instance segmentation and object detection improvements of AP^s :4.7%, AP^s: 4.3% and AP^d: 3.3.%, and AP^d :4.9%, respectively. Achieving such a significant improvement proves that our proposed LPM is indeed able to better exploit the texture detail information of the instances from the original image, and such a prompt is also beneficial for instance segmentation in RSIs. In short, our proposed LPM is effective to address the problem of remote sensing instance segmentation.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:03:10 UTC from IEEE Xplore. Restrictions apply.

2) Effect of Global Prompt Module: The GPM is designed to exploit global contextual information to the target instances from the undownsampled original image, which in turn improves the model's performance for instance segmentation in the form of prompt cues. The third row of Table II shows the performance of the model with the GPM embedded. Compared with the baseline model in the first row, our proposed GPM improves AP^s: 2.2%, AP^s_{50}: 2.6%, AP^s_{75}: 4.3% in instance segmentation; and AP^d: 1.7%, AP^d_{50}: 1.4%, AP^d_{75}: 3.3% in object detection. The third column of Fig. 7 shows the model performance of fine-grained categories with the GPM embedded. Compared with the baseline model in the first column, our proposed GPM achieves an improvement of AP^s: 8.7%, AP^d:8.2% and AP^s: 4.0%, AP^d: 5.2% in the SH and HA categories, respectively, and in most of the other categories to varying degrees. From the analysis, our proposed GPM contains global-to-local prompt paradigm ideas and designed implementation modules that bring different levels of enhancement in overall data and fine-grained categories. In short, our proposed GPM is effective for RSI instance segmentation.

3) Effect of Proposal's Area Loss Function: The PAreaLoss is proposed to add a scale decoupling dimension for the extraction of proposals to achieve more accurate proposals, which in turn exploits the potential of our proposed LPM and GPM to achieve better performance. Compared with the baseline model in the first row of Table II, the fourth row with the addition of PAreaLoss has an improvement in instance segmentation performance: AP^s: 1.6%, AP^s_{50}: 0.9%, AP^s_{75}: 2.3%, and an improvement in object detection performance: AP^d: 1.6%, AP^d_{50}: 0.7%, AP^d_{75}: 2.4%. Compared with the baseline (first row), our proposed three components (fifth row) achieve an improvement in instance segmentation: AP^s: 2.7%, AP^s_{50}: 3.8%, and AP^s: 2.7%, AP^s_{50}: 3.8%, and AP^s_{75}: 4.2% and an improvement in object detection: AP^d: 3.2%, AP^d_{50}: 3.2%, and AP^d_{75}: 4.7%. In the ablation of the fine-grained categories in Fig. 7, PAreaLoss improves to varying degrees in most of the categories, especially for SBF, instance segmentation performance improves by AP^s: 4.1%, and object detection performance improves by AP^d: 4.4%. From the analyses, our proposed PAreaLoss indeed improves the performance of instance segmentation due to its benefit of more accurate proposals, which in turn exploits the potential of LPM and GPM. In short, our proposed PAreaLoss is effective for RSI instance segmentation.

4) Hyperparameter of Prompt Roi Size S_{ps}: As shown in Fig. 8, the ablation experiments of the hyperparameter prompt roi size S_{ps} in the local prompt module are conducted. This hyperparameter determines the uniform size of the target instances after cropping the corresponding image regions in the original image. If this size is too small, it will not be able to fully exploit the texture detail information of the target instances in the original image, and too large it brings redundant computational consumption, so we set up an ablation experiment of this hyperparameter to determine an appropriate value. From the change in the curves in Fig. 8, this size before 28, with the increase in the size, the accuracy of both detection AP^d and segmentation AP^s continues to increase; after more than 28, the two accuracy metrics do not have a significant change, so we determine the value of the hyperparameter S_{ps} as 28.

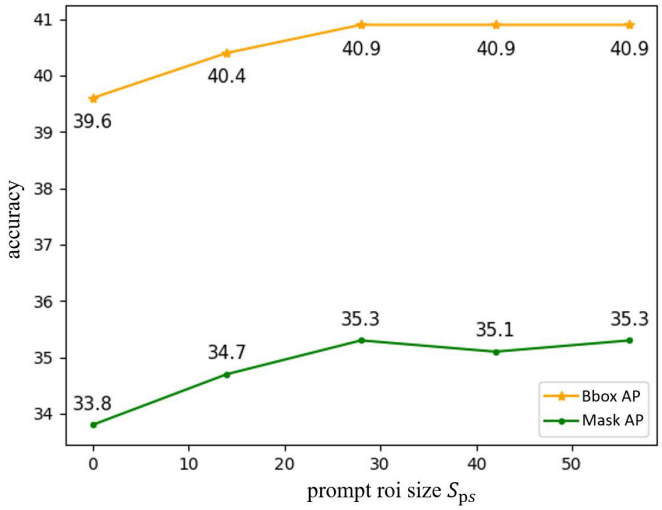


Fig. 8. Ablation of the hyperparameter prompt roi size S_{ps}.

5) Space and Time Complexity Analysis: The space and time complexity comparisons of our proposed approach are shown in the last two columns of Table II. We use the number of model parameters as model size to illustrate model space complexity, as shown in the last column of Table II. We adopt inference speed frames per second (FPS) to illustrate the time complexity of our proposed approach, as shown in the second-last column of Table II. Although our proposed components bring a little parameter and inference time improvement, it is still able to satisfy the needs of instance segmentation applications, and the improvement of instance segmentation performance is more prominent, and therefore, our proposed method is worthwhile.

In conclusion, each component of our proposed prompt paradigm is effective for remote sensing instance segmentation.

D. Qualitative Comparison

For instance segmentation of optical RSIs, the comparison of visualization is shown in Fig. 9. What can be observed from the comparisons is that our proposed approach has several significant improvements over the baseline model Mask-RCNN and the existing sample consistency network (SCNet), including better mask integrity, fewer false alarms, and more accurate mask boundaries. For instance segmentation of SAR images, the visualization results are shown in Fig. 10. From visualization, it can be observed that for the SH samples in SAR images, our proposed approach performs well for a wide range of scenarios such as different scales and complex backgrounds, including near-shore or far-shore. Moreover, the segmentation boundary is accurate compared with the ground truth. From the above qualitative results, it is clear that our proposed prompt paradigm is a competitive approach for instance segmentation of RSIs, both optical and SAR images.

E. Quantitative Comparison

In this section, we conduct a quantitative comparison to other state-of-the-art works on four benchmark datasets:

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:03:10 UTC from IEEE Xplore. Restrictions apply.

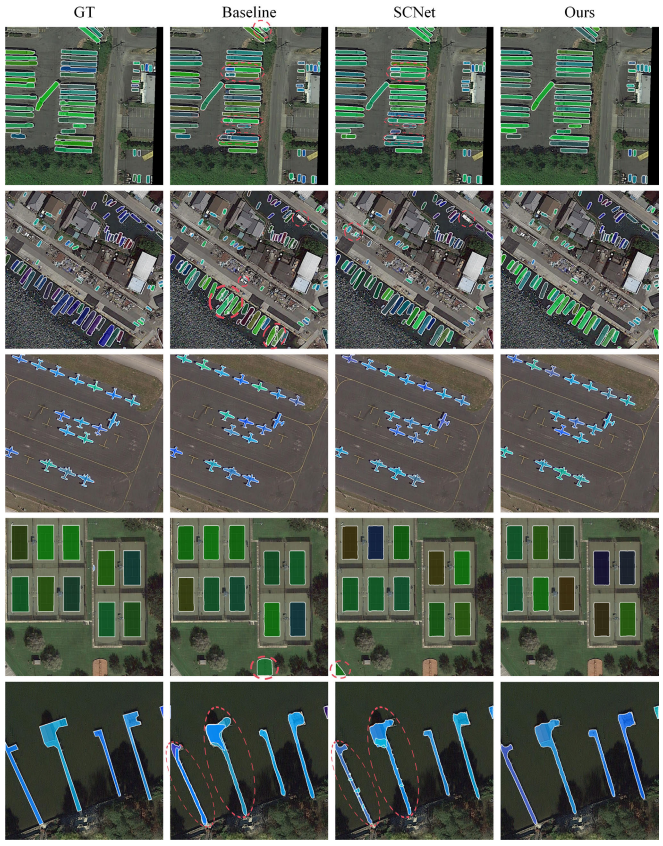


Fig. 9. Visualization results of optical image instance segmentation on the ISAID dataset. The difference is shown in the red dotted circle, and our proposed method is a bit better in terms of the completeness of the segmentation results and so on.

ISAID, NWPU VHR-10, SSDD, and HRSID to illustrate that our proposed approach is a competitive instance segmentation model for RSIs.

1) Instance Segmentation Performance:

a) ISAID: Table III lists the instance segmentation comparison to other advanced works on the ISAID dataset. The object detection performance is shown in Table IV. We conduct experiments by embedding our proposed prompt paradigm into three models: mask RCNN [6], cascade mask RCNN [19], and SCNet [33]. The experimental results based on these three models show that our proposed approach all lead to an improvement in instance segmentation performance, by 2.7% on Mask-RCNN, 2.3% on cascade mask RCNN, and 0.7% on SCNet. The best performance SCNet embedded with our approach outperforms other advanced works in AP, AP_{50}, AP_{75}, and AP_{75}, and AP_s metrics. And the cascade mask RCNN embedded with our approach outperforms other advanced works in AP_m evaluation metrics. Compared with several recent models quad-tree and point rendering network (QTPR-Net) [42], box to mask (Box2Mask) [43], and discrete cosine transform contour (DCTC) [44], our proposed method shows a significant advantage in instance segmentation accuracy, maintaining higher accuracy in all metrics. From the above comparative analysis, it can be concluded that our proposed approach is a competitive RSI instance segmentation work on the ISAID dataset.

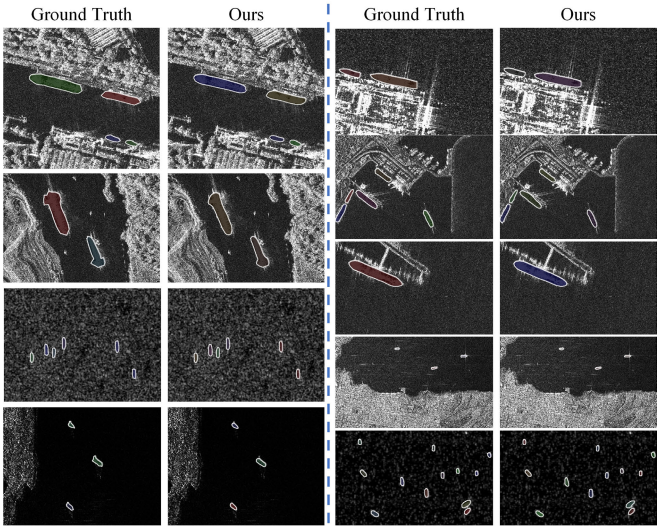


Fig. 10. Visualization results of SAR image instance segmentation on the SSDD dataset. Compared with the ground truth, our proposed method is able to achieve better instance segmentation of the target objects.

b) NWPU VHR-10: The results of the comparison with other works on the NWPU VHR-10 dataset are presented in Table V. The object detection performance is shown in Table VI. According to the comparative results in Table V, the cascade mask RCNN [19] embedded with our proposed approach significantly outperforms the other works, both in terms of the evaluation metrics AP reflected in the performance on the overall data and AP_s reflected in the performance on the small instance set, AP_m on the medium instance set, and AP_l on the large instance set. It can be concluded that our proposed prompt prompt paradigm is a new competitive instance segmentation work on the NWPU VHR-10 dataset.

c) SSDD: Table VII lists the comparison of our proposed approach with other works on the SSDD dataset. And Table VIII lists the object detection performance. Compared with other advanced works, cascade mask RCNN embedded in our proposed prompt paradigm outperforms the performance evaluation metric AP on the overall dataset and AP_m on the medium instance set. For the evaluation metric AP_s on the small instance set, although slightly lower than the last advanced work RSIISN [13], it is significantly higher than the other works. It can be concluded that our proposed prompt paradigm is a competitive instance segmentation work for remote sensing SAR images on the SSDD dataset.

d) HRSID: Table IX lists the comparison of our proposed approach with other works on the HRSID dataset. And Table X lists the object detection performance. From the comparison results, it can be observed that the cascade mask RCNN embedded in our proposed prompt paradigm significantly outperforms other state-of-the-art works. On the overall dataset, the performance improvement over the advanced work RSIISN [13] is AP: 3.5%; on the small instance set, the performance improvement is AP_s: 4.0%. While slightly lower than RSIISN on the medium and large target sets, it is still significantly higher than the other work. Taken together, our proposed prompt paradigm is a competitive remote sensing SAR image instance segmentation work on the HRSID dataset.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:03:10 UTC from IEEE Xplore. Restrictions apply.

TABLE III
QUANTITATIVE INSTANCE SEGMENTATION PERFORMANCE (MASK AP) ON THE ISAID DATASET. # MEANS THE BACKBONE IS ResNet101. "-" MEANS THAT THE VALUE IS NOT REPORTED IN THIS ARTICLE

Model	AP	AP_{50}	AP_{75}	AP_s	AP_m	AP_l
YOLACT [36]	22.3	43.3	19.1	8.4	31.8	40.4
Box2Mask-C# [37]	26.6	50.6	23.8	10.6	33.6	47.4
Ref. [38]	29.4	54.5	27.8	15.5	37.8	42.0
Mask-RCNN [6]	33.8	56.6	35.7	19.4	40.1	45.6
Mask Scoring RCNN [39]	35.9	57.7	38.4	20.8	44.3	51.5
PointRend [40]	35.6	59.0	37.3	20.3	44.8	52.9
Cascade Mask RCNN [19]	35.6	57.8	38.0	20.8	44.3	52.7
HTC [41]	37.4	60.2	40.1	23.5	44.6	53.5
QTPR-Net [42]	37.1	-	-	-	-	-
Box2Mask [43]	34.2	57.5	36.2	19.6	41.4	47.9
DCTC [44]	36.3	58.2	38.0	21.2	45.2	47.3
SCNet [33]	37.3	59.5	40.3	23.3	44.8	52.3
RSIISN [13]	37.6	60.8	41.4	22.4	**47.7**	53.4
Mask RCNN(ours)	36.5	60.4	39.9	21.1	44.1	52.2
Cascade Mask RCNN(ours)	37.9	61.3	40.3	22.0	44.0	**54.9**
SCNet(ours)	**38.0**	**63.3**	**41.5**	**23.8**	45.7	54.3

2) Object Detection Performance: The facilitation between different tasks plays a crucial role in optimizing the learning of the model, e.g., the classification task provides instance category information that facilitates the accuracy of instance segmentation, while instance segmentation facilitates object detection in spatial localization. Therefore, we establish a study on the performance of our proposed approach for the object detection task. After examining the comparison results listed in Tables IV, VI, VIII, and X, what can be found is that our proposed approach outperforms the previous works in object detection on the two datasets except the ISAID and SSDD datasets. From the comparison results in Table IV, although our proposed approach is not optimal, it outperforms all other works except RSIISN, and in addition, the segmentation performance on the ISAID dataset outperforms RSIISN and other works. From the comparison results in Table VIII, although our approach is not optimal in terms of AP evaluation metrics, it is superior to other works in terms of AP_s for small object sets and AP_l metrics for medium object sets. Compared with the latest model QTPR-Net, our proposed method shows significant advantages in object detection accuracy, both in overall performance AP and performance AP and performance AP_s, AP_m, and AP_l, at each scale. The above analyses show that our proposed approach has a positive impact on object detection task in RSIs. The synergistic facilitation between these interrelated tasks helps improve the model to address complex visual problems in RSIs.

To sum up, our proposed prompt paradigm is a competitive approach for instance segmentation of RSIs, both in optical remote sensing data and SAR remote sensing data.

TABLE IV
QUANTITATIVE OBJECT DETECTION PERFORMANCE (BBOX AP) ON THE ISAID DATASET. "-" MEANS THAT THE VALUE IS NOT REPORTED IN THIS ARTICLE

Model	AP	AP_s	AP_m	AP_l
YOLACT [36]	28.8	16.7	33.7	39.7
Mask-RCNN [6]	39.6	25.1	46.4	47.7
PointRend [40]	40.2	26.2	48.0	52.1
Mask Scoring RCNN [39]	40.5	25.8	48.8	53.3
Ref. [38]	-	27.4	47.1	31.1
Cascade Mask RCNN [19]	43.1	28.1	48.7	56.0
RSIISN [13]	**46.6**	**32.3**	**54.8**	**61.6**
QTPR-Net [42]	42.5	-	-	-
Mask-RCNN(ours)	42.8	27.5	49.4	56.4
SCNet(ours)	44.4	29.7	51.0	56.6
Cascade Mask RCNN(ours)	44.8	29.1	49.7	60.0

TABLE V
QUANTITATIVE INSTANCE SEGMENTATION PERFORMANCE (MASK AP) ON THE TEST SET OF THE NWPU VHR-10 DATASET. "-" MEANS THAT THE VALUE IS NOT REPORTED IN THIS ARTICLE

Model	AP	AP_s	AP_m	AP_l
YOLACT [36]	45.2	18.9	44.0	63.1
Mask-RCNN [6]	54.9	33.7	55.1	52.3
PointRend [40]	59.9	46.3	60.5	54.0
Mask Scoring RCNN [39]	57.0	46.7	56.9	55.8
Cascade Mask RCNN [19]	58.1	44.2	58.4	47.1
HRNetV2p-W32 [45]	65.1	49.5	54.7	69.8
FB-ISNet [46]	66.5	-	-	-
Cascade Mask RCNN(ours)	**66.9**	**53.6**	**65.9**	**77.3**

TABLE VI
QUANTITATIVE OBJECT DETECTION PERFORMANCE (BBOX AP) ON THE TEST SET OF THE NWPU VHR-10 DATASET. "-" MEANS THAT THE VALUE IS NOT REPORTED IN THIS ARTICLE

Model	AP	AP_s	AP_m	AP_l
YOLACT [36]	52.7	51.8	52.4	50.8
Mask-RCNN [6]	62.7	66.3	64.4	53.6
PointRend [40]	59.7	61.7	60.6	45.2
Mask Scoring RCNN [39]	61.9	57.9	62.6	48.7
Ref. [38]	53.2	-	-	-
Cascade Mask RCNN [19]	65.7	56.6	66.7	52.4
WSODet [47]	61.3	-	-	-
DAL [48]	63.9	-	-	-
RSIISN [13]	67.6	64.9	68.4	55.9
Cascade Mask RCNN(ours)	**71.3**	**71.1**	**71.6**	**67.7**

F. Instance-Specific Boxes Prompt for Promptable Instance Segmentation

The above experimental sections demonstrate that our proposed approach is beneficial for the remote sensing instance segmentation task. It is worth mentioning that our proposed

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:03:10 UTC from IEEE Xplore. Restrictions apply.

TABLE VII
QUANTITATIVE INSTANCE SEGMENTATION PERFORMANCE (MASK AP) ON THE TEST SET OF THE SSDD DATASET

Model	AP	AP_s	AP_m	AP_l
YOLACT [36]	56.3	57.9	53.6	17.6
FL-CSE-ROIE [49]	62.6	63.3	61.2	75.0
Mask-RCNN [6]	64.2	66.2	58.3	27.6
MAI-SE-Net [50]	63.0	63.3	62.5	47.7
PointRend [40]	65.4	67.4	59.8	39.5
Mask Scoring RCNN [39]	64.6	66.9	57.6	15.0
Cascade Mask RCNN [19]	64.9	67.4	57.5	38.7
C-SE Mask RCNN [51]	58.6	58.3	60.7	26.7
EMIN [52]	61.7	62.1	61.3	40.1
RSIISN [13]	68.4	**69.8**	63.7	**41.0**
Cascade Mask RCNN(ours)	**70.0**	69.1	**73.4**	33.6

TABLE VIII
QUANTITATIVE OBJECT DETECTION PERFORMANCE (BBOX AP) ON THE TEST SET OF THE SSDD DATASET

Model	AP	AP_s	AP_m	AP_l
YOLACT [36]	60.4	64.5	49.8	23.7
Mask-RCNN [6]	68.6	70.9	61.6	8.9
PointRend [40]	67.0	70.1	57.2	13.9
Mask Scoring RCNN [39]	**84.5**	71.0	61.5	7.9
Ref. [38]	66.4	64.3	70.2	**72.0**
Cascade Mask RCNN [19]	69.1	71.4	62.8	9.2
RSIISN [13]	70.3	72.7	62.9	33.9
Cascade Mask RCNN(ours)	73.6	**72.8**	**77.0**	42.7

TABLE IX
QUANTITATIVE INSTANCE SEGMENTATION PERFORMANCE (MASK AP) ON THE HRSID DATASET

Model	AP	AP_s	AP_m	AP_l
YOLACT [36]	31.8	30.0	50.1	13.6
Mask-RCNN [6]	53.4	52.9	62.2	15.6
PointRend [40]	51.8	51.1	61.5	19.6
Mask Scoring RCNN [39]	53.5	53.1	61.6	18.1
Cascade Mask RCNN [19]	54.2	53.7	62.4	22.2
RSIISN [13]	55.8	54.8	**66.7**	**35.2**
Cascade Mask RCNN(ours)	**59.3**	**58.8**	66.2	33.7

TABLE X
QUANTITATIVE OBJECT DETECTION PERFORMANCE (BBOX AP) ON THE HRSID DATASET

Model	AP	AP_s	AP_m	AP_l
YOLACT [36]	53.2	54.8	53.5	21.2
Mask-RCNN [6]	60.9	62.0	64.7	12.9
PointRend [40]	57.7	58.8	60.2	11.7
Mask Scoring RCNN [39]	60.1	61.3	64.5	12.8
Ref. [38]	55.2	38.9	70.2	**65.0**
Cascade Mask RCNN [19]	64.4	65.3	66.7	21.6
RSIISN [13]	63.9	64.4	68.4	36.9
Cascade Mask RCNN(ours)	**70.4**	**71.5**	**71.4**	43.4

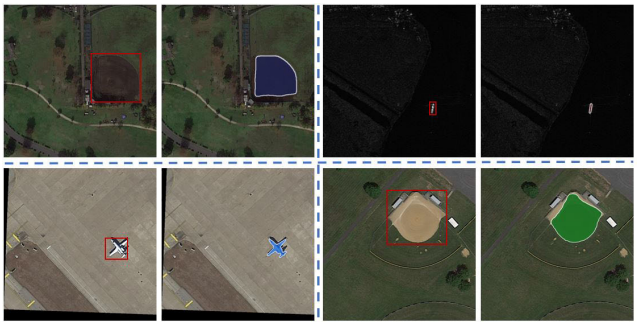


Fig. 11. Examples of promptable instance segmentation. On the left is the given boxes' prompt, and on the right is the corresponding mask results.

TABLE XI
QUANTITATIVE INSTANCE SEGMENTATION PERFORMANCE (MASK AP) ABOUT PROMPTABLE INSTANCE SEGMENTATION. "IS" INDICATES INSTANCE SEGMENTATION

Model	Baseline	IS	Promptable IS
Mask AP	58.1	66.9	67.7

approach supports not only automatic instance segmentation but also promptable instance segmentation with boxes' prompt. Our proposed prompt paradigm enables existing instance segmentation models to be extended to support the promptable instance segmentation task. This section evaluates the performance of our proposed promptable instance segmentation from both the qualitative and quantitative perspectives.

The qualitative promptable instance segmentation with the boxes' prompt is shown in Fig. 11. Given the boxes' prompt, our promptable instance segmentation model can complete the corresponding instance segmentation. First, the model completes the feature extraction of the image encoder based on the input image, and then, based on the input boxes' prompt, the prompt encoder together with the instance segmentation decoder completes the instance segmentation of the corresponding regions. It is worth mentioning that after completing the extraction of the encoder, the promptable instance segmentation in the boxes' prompt takes only 40 ms.

The quantitative promptable instance segmentation evaluation results are shown in Table XI. For some missed instances of automatic instance segmentation, we input the boxes' prompt into the promptable instance segmentation model, output the mask, and finally compute the evaluation metrics together with the automatic instance segmentation results. The reason for few improvements is that the number of missed instances in the automatic instance segmentation model with our prompt paradigm is less compared with the number of recalled instances, but it is enough to prove that the promptable instance segmentation model constructed on the basis of our proposed prompt paradigm is effective.

The above analyses demonstrate that our proposed prompt paradigm can be extended to promptable instance segmentation models with some effectiveness.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:03:10 UTC from IEEE Xplore. Restrictions apply.

V. Conclusion

In this article, we propose a novel prompt paradigm for the problem of unbalanced foreground and background pixel ratios and limited instance sizes in RSIs, as well as the harm of downsampling in the existing deep feature extraction paradigms to address this problem. The local prompt module fully exploits the texture information of the instances in the input image as a local prompt, and the global-to-local prompt module models the contextual information from the global to the local part of the instances as a global prompt, while a constrained loss function on the scale of the proposals is used to better exploit the potential of the two prompt modules. Thorough experiments demonstrate that our proposed approach is effective and competitive for RSI instance segmentation. In future research, we expect to boost the generalization and versatility of promptable RSI instance segmentation models by constructing more appropriate training strategies on large-scale datasets, such as how to use the prompt function to eliminate the disadvantages of the lack of large-scale annotation, or the optimization of the training approach under large-scale data, but this is also beyond the scope of this article.

References

[1] P. Jia, C. Chen, D. Zhang, Y. Sang, and L. Zhang, "Semantic segmentation of deep learning remote sensing images based on band combination principle: Application in urban planning and land use," *Comput. Commun.*, vol. 217, pp. 97–106, Mar. 2024.

[2] R. Zou, J. Liu, H. Pan, D. Tang, and R. Zhou, "An improved instance segmentation method for fast assessment of damaged buildings based on post-earthquake UAV images," *Sensors*, vol. 24, no. 13, p. 4371, Jul. 2024.

[3] J. Rane, O. Kaya, S. K. Mallick, and N. L. Rane, "Artificial intelligence-powered spatial analysis and chatgpt-driven interpretation of remote sensing and gis data," in *Generative Artificial Intelligence in Agriculture, Education, and Business*. London, U.K.: Deep Science Publishing, 2024, pp. 162–217.

[4] F. Chen et al., "Mapping center pivot irrigation systems in global arid regions using instance segmentation and analyzing their spatial relationship with freshwater resources," *Remote Sens. Environ.*, vol. 297, Nov. 2023, Art. no. 113760.

[5] T. Lin et al., "Microsoft COCO: Common objects in context," in *Proc. 13th Eur. Conf. Comput. Vis. (ECCV)*. Zurich, Switzerland: Springer, 2014, pp. 740–755.

[6] K. He, G. Gkioxari, P. Dollár, and R. Girshick, "Mask R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2961–2969.

[7] S. Liu, L. Qi, H. Qin, J. Shi, and J. Jia, "Path aggregation network for instance segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 8759–8768.

[8] H. Chen, K. Sun, Z. Tian, C. Shen, Y. Huang, and Y. Yan, "BlendMask: Top-down meets bottom-up for instance segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, May 2020, pp. 8573–8581.

[9] S. W. Zamir et al., "ISAID: A large-scale dataset for instance segmentation in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops*, Jul. 2019, pp. 28–37.

[10] G. Cheng, P. Zhou, and J. Han, "Learning rotation-invariant convolutional neural networks for object detection in VHR optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 12, pp. 7405–7415, Dec. 2016.

[11] Z. Chen, Y. Shang, A. Python, Y. Cai, and J. Yin, "DB-BlendMask: Decomposed attention and balanced blendmask for instance segmentation of high-resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5615915.

[12] T. Zhang et al., "Semantic attention and scale complementary network for instance segmentation in remote sensing images," *IEEE Trans. Cybern.*, vol. 52, no. 10, pp. 10999–11013, Oct. 2022.

[13] W. Ye, W. Zhang, W. Lei, W. Zhang, X. Chen, and Y. Wang, "Remote sensing image instance segmentation network with transformer and multi-scale feature representation," *Expert Syst. Appl.*, vol. 234, Dec. 2023, Art. no. 121007.

[14] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[15] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022.

[16] A. Kirillov et al., "Segment anything," 2023, *arXiv:2304.02643*.

[17] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[18] L. Mou and X. X. Zhu, "Vehicle instance segmentation from aerial image and video using a multitask learning residual fully convolutional network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 11, pp. 6699–6711, Nov. 2018.

[19] Z. Cai and N. Vasconcelos, "Cascade R-CNN: Delving into high quality object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6154–6162.

[20] J. Wang, S. Ji, and T. Zhang, "From image transfer to object transfer: Cross-domain instance segmentation based on center point feature alignment," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4407011.

[21] K. Sun, B. Xiao, D. Liu, and J. Wang, "Deep high-resolution representation learning for human pose estimation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Mar. 2019, pp. 5693–5703.

[22] M. Jia et al., "Visual prompt tuning," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, Oct. 2022, pp. 709–727.

[23] P. Liu, "Pre-train, prompt, and predict: A systematic survey of prompting methods in natural language processing," *ACM Comput. Surv.*, vol. 55, no. 9, pp. 1–35, 2023.

[24] K. Zhou, J. Yang, C. C. Loy, and Z. Liu, "Learning to prompt for vision-language models," *Int. J. Comput. Vis.*, vol. 130, no. 9, pp. 2337–2348, Sep. 2022.

[25] C. Ruan and H. Wang, "Dynamic visual prompt tuning for parameter efficient transfer learning," 2023, *arXiv:2309.06123*.

[26] K. Chen et al., "RSPrompter: Learning to prompt for remote sensing instance segmentation based on visual foundation model," 2023, *arXiv:2306.16269*.

[27] Y. Liu et al., "A survey of visual transformers," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 6, pp. 7478–7498, Jun. 2024.

[28] J. Yang et al., "Focal self-attention for local–global interactions in vision transformers," 2021, *arXiv:2107.00641*.

[29] S. Ren, X. Yang, S. Liu, and X. Wang, "SG-former: Self-guided transformer with evolving token reallocation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 6003–6014.

[30] R. Ronen, S. Tsiper, O. Anschel, I. Lavi, A. Markovitz, and R. Manmatha, "GLASS: Global to local attention for scene-text spotting," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*. Cham, Switzerland: Springer, 2022, pp. 249–266.

[31] T. Zhang et al., "SAR ship detection dataset (SSDD): Official release and comprehensive data analysis," *Remote Sens.*, vol. 13, no. 18, p. 3690, Sep. 2021.

[32] S. Wei, X. Zeng, Q. Qu, M. Wang, H. Su, and J. Shi, "HRSID: A high-resolution SAR images dataset for ship detection and instance segmentation," *IEEE Access*, vol. 8, pp. 120234–120254, 2020.

[33] T. Vu, H. Kang, and C. D. Yoo, "SCNet: Training inference sample consistency for instance segmentation," in *Proc. AAAI Conf. Artif. Intell.*, Feb. 2021, pp. 2701–2709.

[34] K. Chen et al., "MMDetection: Open MMLab detection toolbox and benchmark," 2019, *arXiv:1906.07155*.

[35] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Nov. 2009, pp. 248–255.

[36] D. Bolya, C. Zhou, F. Xiao, and Y. J. Lee, "YOLACT: Real-time instance segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9157–9166.

[37] W. Li et al., "Box2Mask: Box-supervised instance segmentation via level-set evolution," 2022, *arXiv:2212.01579*.

[38] Y. Luo, J. Han, Z. Liu, M. Wang, and G.-S. Xia, "An elliptic centerness for object instance segmentation in aerial images," *J. Remote Sens.*, vol. 2022, pp. 1–13, Jan. 2022.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:03:10 UTC from IEEE Xplore. Restrictions apply.

[39] Z. Huang, L. Huang, Y. Gong, C. Huang, and X. Wang, "Mask scoring R-CNN," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 6409–6418.
[40] A. Kirillov, Y. Wu, K. He, and R. Girshick, "PointRend: Image segmentation as rendering," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9799–9808.
[41] K. Chen et al., "Hybrid task cascade for instance segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 4974–4983.
[42] X. Zhang, J. Shen, H. Hu, and H. Yang, "A new instance segmentation model for high-resolution remote sensing images based on edge processing," *Mathematics*, vol. 12, no. 18, p. 2905, Sep. 2024.
[43] W. Li et al., "Box2Mask: Box-supervised instance segmentation via level-set evolution," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 7, pp. 5157–5173, Jul. 2024.
[44] Z. Chen, T. Liu, X. Xu, J. Leng, and Z. Chen, "DCTC: Fast and accurate contour-based instance segmentation with DCT encoding for high-resolution remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 8697–8709, 2024.
[45] D. Kumar, "Accurate object detection & instance segmentation of remote sensing, imagery using cascade mask R-CNN with hrnet backbone," 2022.
[46] H. Su, P. Huang, J. Yin, and X. Zhang, "Faster and better instance segmentation for large scene remote sensing imagery," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2022, pp. 2187–2190.
[47] Z. Tan, Z. Jiang, C. Guo, and H. Zhang, "WSODet: A weakly supervised oriented detector for aerial object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5604012.
[48] X. Zhang, H. Luo, S. Zhong, L. Tang, J. Peng, and J. Fan, "Movable object detection in remote sensing images via dynamic automatic learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 2001314.
[49] T. Zhang and X. Zhang, "A full-level context squeeze-and-excitation ROI extractor for SAR ship instance segmentation," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[50] T. Zhang and X. Zhang, "A mask attention interaction and scale enhancement network for SAR ship instance segmentation," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[51] T. Zhang, X. Zhang, J. Li, and J. Shi, "Contextual squeeze-and-excitation mask R-CNN for SAR ship instance segmentation," in *Proc. IEEE Radar Conf.*, Mar. 2022, pp. 1–6.
[52] T. Zhang and X. Zhang, "Enhanced mask interaction network for SAR ship instance segmentation," in *Proc. IGARSS - IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 3508–3511.



Xuexue Li (Student Member, IEEE) received the B.Sc. degree from the University of Science and Technology Beijing, Beijing, China, in 2020. He is currently pursuing the Ph.D. degree with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing. His research interests include computer vision, pattern recognition, and remote sensing and UAV image processing, especially on object detection.



Wenhui Diao (Member, IEEE) received the B.Sc. degree from Xidian University, Xi'an, China, in 2011, and the M.Sc. and Ph.D. degrees from the Institute of Electronics, Chinese Academy of Sciences, Beijing, China, in 2016.

He is currently an Assistant Professor with the Institute of Electronics, Chinese Academy of Sciences. His research interests include computer vision and remote sensing image analysis.



Xinming Li received the B.Sc. degree from Hunan University, Changsha, China, in 1985, and the Ph.D. degree from Nanjing University of Science and Technology, Nanjing, China, in 2010.

He is currently a Professor with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China. His research interests include edge information processing and intelligent service.



Xian Sun (Senior Member, IEEE) received the B.Sc. degree from Beijing University of Aeronautics, Beijing, China, in 2004, and the M.Sc. and Ph.D. degrees from the Institute of Electronics, Chinese Academy of Sciences, Beijing, in 2009.

He is currently a Professor with the Aerospace Information Research Institute, Chinese Academy of Sciences. His research interests include computer vision, geospatial data mining, and remote sensing image understanding.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:03:10 UTC from IEEE Xplore. Restrictions apply.